ARGUS: final technical disclosure and evidence

Implemented reference-selection method, bounded synthetic evidence, draft claim concepts and enabling drawings.

Read this first

The optional selector has positive evidence for diagnostic-arrival prediction in a declared scalar simulation. Earlier waveform stress failures remain visible. The final package identifies the implemented technical interaction, claim-support limits and physical-acquisition requirements; novelty, inventorship and filing status remain external questions.

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Source documents and exact commands remain in docs/. Current JSON and JUnit evidence remain in output/release/ and research_results/. The separate manuscript PDF is compiled from paper/main.tex.

Drawings and physical boundary

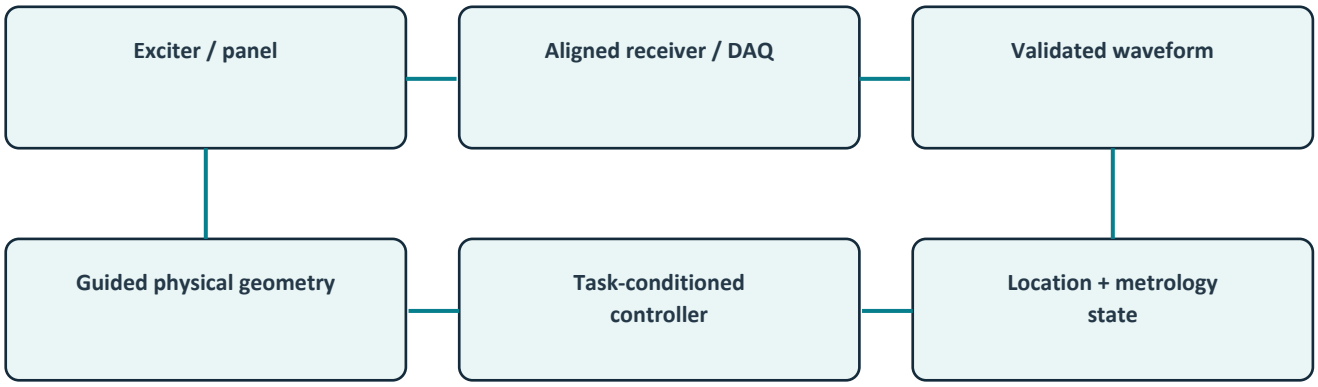


Figure 1. Control-system blocks; the aligned physical DAQ is a proposed apparatus embodiment requiring measured timing.

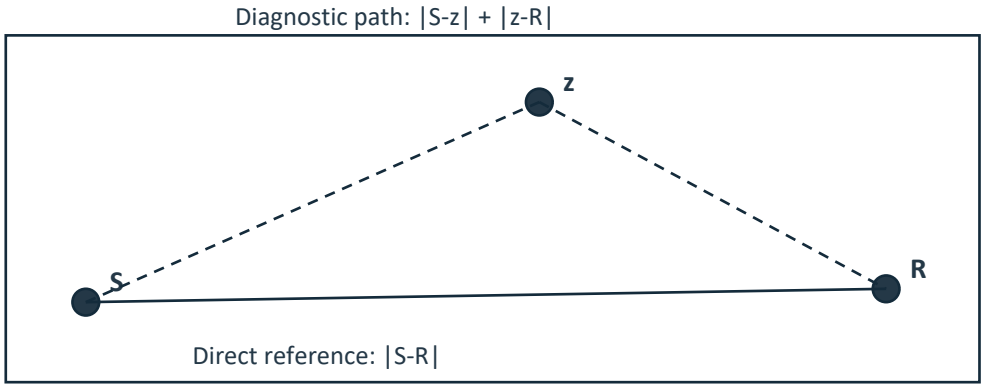


Figure 2. Direct-reference and bistatic diagnostic geometry on a rectangular panel.

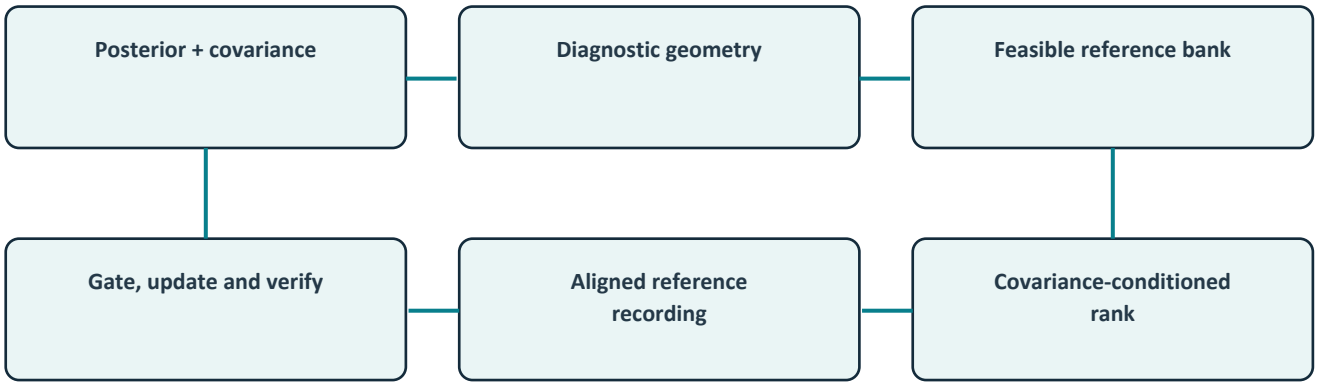


Figure 3. Implemented optional reference selection and gated update sequence. Covariance ranking uses all posterior mass.

ARGUS final technical disclosure for patent review

Revision: final technical-review version, 7 October 2026. **Status:** implemented software and synthetic evidence; no patent filing, novelty opinion, physical apparatus validation, or human inventorship finding.

Title and supported technical contribution

Task-conditioned reference geometry selection for sequential acoustic interrogation with correlated propagation uncertainty and gated verification.

The narrower candidate contribution is an acoustic control sequence that uses the current location probability map to choose which propagation reference to measure before the next diagnostic experiment. It reduces uncertainty relevant to that diagnostic geometry, rather than selecting a reference merely because it is first feasible or reducing every instrument parameter indiscriminately.

The new full-grid reference selector is implemented and exercised in the live engine. It is an optional fast-controller extension. Established Bayesian conditioning and experimental-design principles are not claimed as inventions. The separate two-observation physical-loss planner already compares calibration and diagnosis under a coarsened arrival model; the new method is a cheaper, full-grid, task-conditioned reference subproblem, not a claim to supersede it.

Technical problem and mechanism

An observed arrival combines propagation speed and electronics timing. A single reference path gives one constraint on two correlated unknowns. Repeating the same path can reduce variance along one observed direction while leaving other diagnostic paths uncertain. Generic uncertainty reduction may improve timing offset while offering little benefit for the intended diagnostic path.

ARGUS keeps the correlated speed/timing covariance, evaluates a finite feasible reference bank against all spatial posterior mass under a nominated diagnostic geometry, predicts the covariance after each reference, and selects the minimum expected propagation-variance-plus-cost objective. The actual recording is then validated and incorporated as metrology evidence. It is not counted as defect-location evidence. Later diagnosis and fresh diverse verification use the updated propagation model.

Executed method and implementation boundaries

1. Maintain a normalized 2D location grid and correlated speed/timing means and covariance. Other nuisance variables remain approximate separate marginals.

2. The existing fast action controller decides whether reference calibration is warranted. The new selector does not independently replace that decision.

3. Form three direct-reference geometries with distinct physical path lengths. Filter no-go regions, unavailable actions, waveform/drive, spacing and frequency restrictions before selection.

4. Nominate a feasible diagnostic geometry from the current baseline candidates. Convert every grid cell to physical coordinates and its bistatic path length.

5. Calculate the weighted second moment of arrival derivatives over the complete grid. Retain speed/timing covariance cross terms; do not discard low-mass cells.

6. For each feasible reference, condition the propagation covariance using the same first-order reference model and declared noise/quality assumptions.

7. Rank by predicted path-equivalent propagation variance plus declared acquisition cost. Report before/after variance, paths, assumptions, design rank, costs and every reference alternative. Rank is diagnostic, not a forced diversity rule.

8. Acquire one supported reference. Validate original bandwidth/window/units and channel quality before updating metrology; block incompatible data.

9. Replan diagnosis from the updated model, gate structural evidence before fusion, and require a fresh qualifying verification before a confident stop.

10. Preserve the action, prediction, observation, actual source identity and state transition in content-bound evidence records.

Activation is `enable_task_conditioned_calibration=true` in the session API or `--task-calibration` in scripts/demo_simulation.py. It stays off by default. The optional two-step risk policy takes precedence if that policy is enabled. No robot motion, high-fidelity FEM switching, depth/type/severity inference or measured safety decision is included in this supported implementation.

Mathematical specification

Let z be a grid-cell center with mass p_z, diagnostic source S and receiver R, effective speed v and timing offset t0. All distances are physical metres.

```
ell_z = distance(S,z) + distance(z,R)
mu_z = t0 + ell_z/v
g_z = [-ell_z/v^2, 1]
W = sum_z p_z * g_z^T g_z
V_before = trace(W C)                          [s^2]
```

For direct-reference length d, h=[-d/v^2,1]. R_ref is the declared timing and placement variance; q=0.75 is the selector's explicitly assumed quality.

```
K = C h^T / (h C h^T + R_ref/q)
C_after = (I-Kh) C (I-Kh)^T + K(R_ref/q)K^T
V_after = trace(W C_after)
reference_objective = v^2 * V_after + c * acquisition_cost
```

Here v^2 V is path-equivalent propagation variance in m^2, not expected location error. The default c=0.0001 m^2 per research cost unit is a preference, not a measured monetary, energy or maintenance loss. The comparison study disables cost to isolate the geometry criterion. Actual acquisition quality may differ from the assumed value and is checked before the measured update.

For a valid Gaussian block, C-C_after is positive semidefinite and the modeled variance cannot increase. The reduction equals

```
sum_z p_z * (g_z C h^T)^2 / (h C h^T + R_ref/q)
```

This is an analytic property of the declared local model, not a physical coverage guarantee. The implementation verifies variance bounds and does not mutate the belief, hidden truth, random generator or covariance during selection.

Concrete software embodiment

The worked configuration uses a 600 x 400 mm panel, a 20 x 20 grid, 16 kHz acquisition, an illustrative 180 m/s speed and 0.8 ms timing prior. Reference paths are 0.18, 0.36 and 0.54 m; the measured noise contract includes a 0.2 ms timing floor and declared pose uncertainty. These are simulation/design values, not certified material or transducer constants.

In the checked easy seed 17 eight-action demonstration, the calibration at action 7 selects a 0.54 m reference. Its modeled downstream arrival variance falls from 1.767e-7 to 5.463e-8 s^2 under the assumed reference model. The actual simulation continues to approximately 10 mm MAP error and STOP_BUDGET; this one run is not an accuracy superiority result.

Physical acquisition embodiment and enabling requirements

The control software can consume an externally acquired, trigger-aligned WAV with known source/receiver geometry, source sampling rate, excitation definition, recording window and normalization. A proposed physical embodiment uses a shared trigger driving a characterized exciter and acquisition system on a repeatable fixture, records before/through early arrival, and retains the actual timing and transfer calibration. Its wiring and apparatus timing must be measured.

The current ESP32 firmware calls excitation before ADC recording and does not carry a verified sample-clock/trigger contract. It also derives a chirp endpoint internally rather than transporting the complete declared experiment. It is a decay-recording example. Its serial adapter is now refused by arrival localization before any actuation. No synchronized ESP32 apparatus, flash, compilation or physical verification is claimed by this disclosure.

For enabling drawings, Figure 1 specifies the controller/data blocks; Figure 2 shows the direct and bistatic paths; Figure 3 shows the reference-design/update sequence. The physical clock requirement is an implementation boundary, not an omitted claim of an already-built transducer or DAQ.

Technical effects and evidence

The fresh scalar-arrival study contains 64 paired seeds, 3 references per policy, and 192 completed policy runs. Mean downstream arrival RMSE is 279.65 microseconds for the cyclic first-feasible policy and 200.57 for task-conditioned selection, a 28.3% reduction. The paired mean advantage is 79.07 microseconds with bootstrap 95% interval [50.27,110.35]. Source hashes were stable and match the reviewed code.

The mean timing-offset error increases from 136.20 to 144.89 microseconds, while speed error decreases from 14.10 to 11.80 m/s. This adverse tradeoff is retained: the objective targets the subsequent arrival prediction, not all parameter errors. The repeated-short policy is an additional failure control, not the existing cyclic baseline. An earlier repeated-path pilot is preserved and labelled pilot.

The result supports a bounded computational calibration effect in the declared synthetic model. It does not establish real waveform performance, hidden-defect localization, fewer physical measurements, or patent novelty. The earlier 48-run defect-localization study belongs to the pre-extension source and still contains large mismatch errors; it must not be reassigned to this source revision.

Draft claim concepts for practitioner revision

These are technical drafting aids. They have not been assessed for novelty, inventive step, eligibility, unity, clarity, ownership or enforceability.

1. A processor-controlled acoustic interrogation method maintaining a probability distribution over candidate positions and a correlated propagation-speed/timing state; determining a feasible diagnostic geometry; evaluating feasible direct references using the position-weighted covariance-conditioned arrival variance for that geometry; selecting and executing a reference under declared cost; validating its recording and updating propagation state; and subsequently acquiring and fusing diagnostic evidence under the updated state.

2. The method of concept 1, wherein every grid-cell mass contributes to a derivative second-moment matrix and covariance cross terms contribute to reference value.

3. The method of concept 1, wherein the direct-reference observation supplies one scalar update to the correlated speed/timing block, with a Joseph covariance update and without multiplication into the structural location grid.

4. The method of concept 1, wherein reference eligibility is tested against physical placement, source bandwidth/window and drive constraints before acquisition.

5. The method of concept 1, wherein reference-path rank/span and before/after target variance are retained without claiming separate identification from one delay.

6. The method of concept 1, wherein actual quality/channel/model-domain checks suppress or temper unsupported evidence before a structural update.

7. The method of concept 1, wherein confident termination requires the latest accepted nominal verification using changed geometry or excitation content; reciprocal-path or amplitude-only changes do not alone qualify.

8. The method of concept 1, wherein evidence binds selected geometry, prediction, recorded waveform, calibration and structural transition to source identity.

9. A control-system implementation having a processor, memory and acquisition adapter configured to perform the supported method concepts; its physical trigger/DAQ embodiment is subject to the stated enabling/validation boundary.

The independent concept should be narrowed to the technical interaction actually supported, after a professional search. Security/UI/hash features alone should not be used to manufacture an inventive acoustic effect.

Closest-art comparison and obviousness risks

Reference reviewed	Relevant claim subject matter	Comparison still required
EP4592907A1, claims 1-3	Digital-twin synchronization with a state-wide Bayesian filter; later weighted instantiations and active instrument adjustments	Compare specific full-grid downstream acoustic-reference criterion and ordered measured fusion; generic Bayesian calibration is already crowded
EP2214009A2, claim 1 and dependent claims	Reference wave responses, decorrelation/error functions and spatial probability mapping	Compare reference-geometry selection for a nominated diagnostic task; baseline wave inference is not new
US10126274B2, claims 1 and 4	Known-state data, movable transducer and stored scattered impulse responses for multi-path detection/localization	Compare task-conditioned direct-reference calibration; movable multi-path acquisition itself is not new
Bayesian experimental design and guided-wave placement literature	Value/information-based measurement and sensor design	Task-conditioned covariance reduction may be viewed as established optimal-design mathematics applied to this apparatus

Sources: EP4592907A1, EP2214009A2, US10126274B2. The correct digital-twin reference is EP4592907A1; a transposed publication number must not be used. This limited claim reading is not an exhaustive family search or a finding that the ordered combination is absent or non-obvious.

Human inventorship, ownership and disclosure record

An actual person must supply claim-specific conception and contribution records. This computer-assisted engineering work does not establish who conceived each claim. Preserve the prompts, dated human notes, alternatives, decisions and source versions without treating automatic code generation as a human invention date. Complete university/credit-project ownership and collaborator agreements, prior public demonstrations/repository history, earliest disclosures and intended jurisdiction.

Current official references include India's notified CRI 2025 guidance (review the actual applicable text), the EPO technical-effect framework, and revised USPTO AI-assisted inventorship guidance. They do not establish ARGUS eligibility. The USPTO 2025 revision superseded its 2024 guidance; only natural persons can be named as inventors under that guidance.

Sources: Indian government CRI announcement, USPTO revised guidance, EPO simulation guidance.

Final review handoff

Provide this disclosure, code mapping, new study, negative tradeoff, drawings, regression report and independently retained source manifest to your supervisor and patent practitioner. Before broader claims, obtain blind physical data with measured timing/geometry and specimen-held-out comparisons. Do not state patent pending, field validated, world first, certified or grant guaranteed without the corresponding actual event and evidence. No filing or external publication was performed by this technical finalization.

Final patent-review evidence and reproducibility

The final technical contribution is documented in ARGUS_FINAL_TECHNICAL_DISCLOSURE. It is an optional task-conditioned reference selector, not a new general Bayesian principle or a validated physical defect-localization upgrade.

Implementation mapping

Element	Source	Actual support
Full-grid target matrix and reference ranking	backend/app/active_learning/reference_design.py	Covariance conditioning, costs, all 400 default cells, alternatives and rank/span
Engine invocation	backend/app/services/engine.py	Direct calibration only, feasible nominated diagnostic, explicit optional flag
Propagation measurement update	backend/app/inference/nuisance_posterior.py	Correlated first-order update; one delay is one observation
API/CLI activation	schemas/api.py; session_manager.py; scripts/demo_simulation.py	Persisted enable_task_conditioned_calibration flag and CLI demonstration
Legacy acquisition boundary	hardware/devices.py; firmware/esp32/README.md	Post-excitation example blocked before localization actuation
Functional regressions	backend/tests/test_reference_design.py	Monotonic/PSD variance, full mass, ordering, costs, immutability, observability and engine use
New comparative study	scripts/validate_task_calibration.py	Three equal-budget policies and source-bound per-case outputs

Evidence levels

1. Mathematical invariants apply to the declared local Gaussian model.

2. Twelve new regression cases passed; the full final suite records 306 passes, one refused-legacy-checkpoint skip, and a nonfatal framework warning.

3. The 64-seed calibration comparison is synthetic scalar-arrival evidence.

4. The existing 48-run hidden-defect evaluation is pre-extension historical evidence.

5. Neither experiment is measured apparatus or field performance.

Calibration policy	Mean target arrival RMSE	Mean velocity absolute error	Mean timing absolute error
Repeated short reference (failure control)	454.43 microseconds	20.06 m/s	121.45 microseconds
Cyclic first-feasible (primary baseline)	279.65 microseconds	14.10 m/s	136.20 microseconds
Task-conditioned reference	200.57 microseconds	11.80 m/s	144.89 microseconds

Primary paired advantage: 79.07 microseconds, 95% bootstrap interval [50.27,110.35]. All 192 runs completed. The worsened timing-parameter error is retained; task prediction improves without claiming every parameter improves. The first repeated-reference pilot is not substituted for the cyclic baseline.

Reproduction and source boundary

```
.venv\Scripts\python.exe -m pytest backend\tests -q -o addopts=""
  --junitxml=output/release/final-backend-tests.xml
.venv\Scripts\python.exe scripts\validate_task_calibration.py
.venv\Scripts\python.exe scripts\demo_simulation.py --task-calibration --preset
  easy --seed 17 --experiments 8
```

The model-study report is research_results/final_task_calibration_2026_10_07.json. Check source_changed_during_run=false and compare every retained backend source hash before reusing its results. Protocol and acquisition assumptions are in FINAL_CALIBRATION_PROTOCOL.md. Preserve failures and complete per-case traces.

The final source/evidence ZIP and its independent manifest retain exact source versions and exclude private environment files, credentials and model binaries. Integrity is not an institutional signature or proof of sensor authenticity. External publication and patent filing remain unperformed.

Final task-conditioned reference-design validation protocol

Recorded before the numerical run. This is a synthetic scalar direct-arrival calibration study, not defect-localization or physical apparatus validation.

• Sixty-four final paired seeds 83000-83063; three references per policy.

• Policies: cyclic first-feasible (existing engine reference order), task-conditioned, and repeated-short reference as a separate identifiability failure control.

• Reference paths 0.18, 0.36, 0.54 m on a 600 x 400 mm panel; same bank and budgets.

• A uniform 20 x 20 posterior and fixed diagnostic geometry define the full-grid propagation-arrival design target. No hidden truth enters selection.

• Hidden speed/timing are seeded draws from the declared prior within its bounds. Scalar observations use t0+d/v plus shared per-step Gaussian innovations.

• Acquisition noise is computed once per case from the sealed true speed and declared timing/pose noise, independent of each policy's inference state. The reference updater uses its current assumed noise contract. All policies receive the same per-step Gaussian noise innovation and acquisition variance.

• Report all runs, selected paths, before/after modeled variance, held-out target arrival error, velocity/timing errors and paired bootstrap intervals.

• Primary outcome: held-out target arrival RMSE across grid cells in microseconds. This is propagation prediction, not defect position or safety probability.

• Do not tune to the results, remove seeds, infer field accuracy, or present covariance conditioning or Bayesian design as newly invented mathematics.

• Hash every backend/app Python source and the validation script before/after.

An earlier 81000-series development pilot used a repeated-short baseline. It is retained as final_task_calibration_pilot_repeated_path.json and must not be represented as the existing engine's rotating first-feasible policy. The corrected comparison uses fresh seeds and reports the cyclic baseline as its primary pair.

The 82000-series development run had estimator-dependent acquisition noise. It is retained as final_task_calibration_pilot_coupled_noise.json and excluded from final claims. The corrected final fixture uses independent acquisition physics and fresh 83000-series seeds; this correction is not policy hyperparameter tuning.